An Introduction to Quantum Walks and their properties

Coloquio del Departamento de Ingeniería Matemática

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Motivation

QWs are Quantum mechanical analogue of classical Random Walks
They model a Quantum particle moving on a grid in **discrete time**

$$ih\frac{d}{dt}\psi(t) = H\psi(t) \longrightarrow \psi(t) = \left(e^{-i\frac{H}{h}}\right)^t \psi_0$$

Discrete time, e.g. look at $t_0 = 0$, $t_1 = 1$,...

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Discrete time, e.g. look at $t_0 = 0$, $t_1 = 1$,...
Applications

- Simple model for dynamics of quantum particle
- Quantum Computing

Definitions

Discrete *grid* Γ (e.g. $\Gamma = \mathbb{Z}^d$) and finite-dimensional *coin space* $\mathbb{C}^s$ (e.g. spin)
$\Longrightarrow$ state of particle is unit $\psi \in l^2(\Gamma) \otimes \mathbb{C}^s$

Definition

One time-step of a QW is given by $U\psi$, where U is a unitary, local operator on $l^2(\Gamma) \otimes \mathbb{C}^s$.

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Shift-coin decomposition:

For $(C_x)_{x\in\Gamma}$ family of unitary matrices, define coin operator $\mathcal{C} = \{C_x\}_{x\in\Gamma}$:

$$\mathcal{C}(|x\rangle \otimes e_i) = |x\rangle \otimes C_x e_i$$

Let $(S_i)_{1\leqslant i\leqslant s}$ be family of shifts of $l^2(\Gamma)$, define the conditional shift operator S:

$$S = \sum_{i=1}^{s} S_i \otimes |e_i\rangle\langle e_i|$$

Then $U = S\mathcal{C}$.

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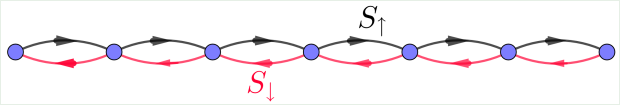
Hadamard coin

Example

Take $\Gamma = \mathbb{Z}$, $s = 2$. Fix ONB $\{|\uparrow\rangle, |\downarrow\rangle\}$ of $\mathbb{C}^2$. The state of the Walker is given by

$$\psi = \begin{pmatrix} \psi_\uparrow \\ \psi_\downarrow \end{pmatrix} \quad \text{s.t. } \|\psi\|_{l^2(\mathbb{Z},\mathbb{C}^2)} = 1$$

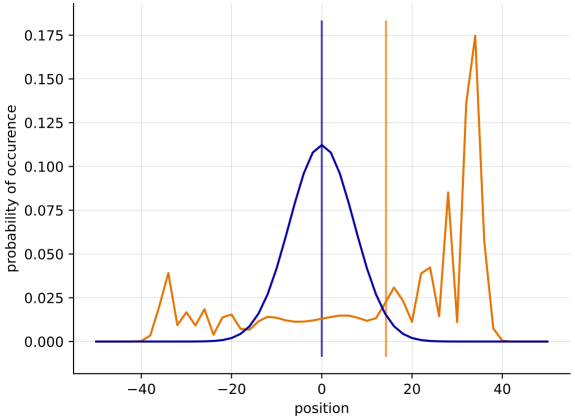
Set $S_\uparrow |x\rangle = |x+1\rangle$ and $S_\downarrow |x\rangle = |x-1\rangle$. Let $C_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.



Note: Measurement after every step gives classical RW with transition probabilities $\frac{1}{2}$.

Difference between classical RW and QW

Hadamard Walk after $N = 50$ steps for $\psi_0 = |0\rangle \otimes |\uparrow\rangle$: Classical RW and QW



- limiting distribution
- transport
- speed of "spreading" ($\sigma = N$ vs $\sigma = \sqrt{N}$)

Relation to Quantum Computing

Intuition: QW spread faster

Grover's algorithm

Database $D = \{0, 1, ..., N-1\}$ with $f : D \to \{0, 1\}$. Search unique $\omega \in D$ with $f(\omega) = 1$.

1. Initialize $\psi_0 = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$
2. Repeat $r(N)$ times:
   - i. apply $U_\omega = \sum_{x=0}^{N-1} (-1)^{f(x)} |x\rangle\langle x|$
   - ii. apply $U_{\psi_0} = 2|\psi_0\rangle\langle\psi_0| - I$
3. measure in basis $(|x\rangle)$

Need $r(N) = \mathcal{O}(\sqrt{N})$ applications

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Relevance of the spectrum

For A self-adjoint, decompose into spectral subspaces

- $\mathcal{H}_p = \overline{\text{span}(\textit{eigenvectors})}$
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RAGE Theorem (Ruelle, Amrein, Georgescu, Enss)

Let K_n be a sequence of relatively compact operators w.r.t. A converging strongly to identity, then:

- $\mathcal{H}_p = \{\psi \in \mathcal{H} \mid \lim_{n\to\infty} \sup_{t\geqslant 0} \|(1-K_n)e^{-itA}\psi\| = 0\}$
- $\mathcal{H}_c = \{\psi \in \mathcal{H} \mid \lim_{n\to\infty} \lim_{T\to\infty} \frac{1}{T}\int_0^T \|K_n\, e^{-itA}\psi\|\, dt = 0\}$

Intuition:
point spectrum $\longrightarrow$ states that stay in some compact set
continuous spectrum $\longrightarrow$ states that leave any compact set

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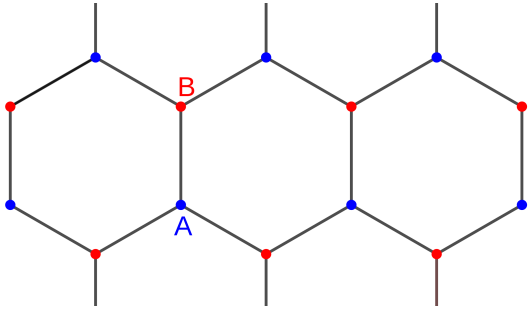
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The hexagonal lattice

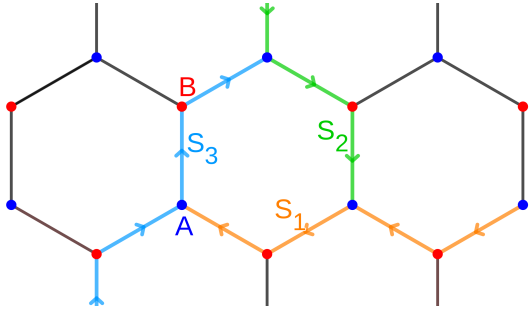
Take Γ to be the hexagonal lattice



Every vertex has degree $3 \longrightarrow s = 3$

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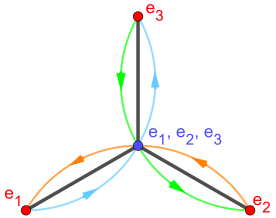
Every vertex has degree 3 $\longrightarrow s = 3$

Define shifts S_1, S_2, S_3 as above, then

$$S = S_1 \otimes |e_1\rangle\langle e_1| + S_2 \otimes |e_2\rangle\langle e_2| + S_3 \otimes |e_3\rangle\langle e_3|$$

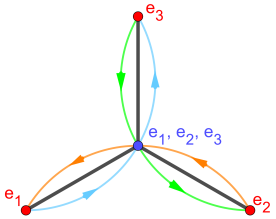
Full Localization

Choosing $C_x = C_0 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ induces the dynamics



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Together with $\tilde{C}_0 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ this is the only coin matrix leading to full localization

$\longrightarrow$ need to change model to obtain localization on open set of coin matrices

Dynamical Localization

Equip every $x \in \Gamma$ with RV ω_x, st $(\omega_x)_{x\in\Gamma}$ is i.i.d. uniformly distributed on $\mathbb{T} = [0, 2\pi)$. Define

$$D_\omega(|x\rangle \otimes \tau) = e^{i\omega_x} |x\rangle \otimes \tau \quad \text{and} \quad U_\omega = D_\omega U$$

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Theorem

There exist $c, g, \varphi > 0$ such that $\sup_{x\in\Gamma} \|C_x - C_0\| \leqslant \varphi$ implies for all $x, y \in \Gamma$ and coin states $\tau, \tau' \in \mathbb{C}^3$

$$\mathbb{E}\left(\sup_{n\in\mathbb{Z}} |\langle x \otimes \tau \mid U_\omega^n\, y \otimes \tau'\rangle|\right) \leqslant c e^{-g\, dist(x,y)}.$$

Probability to move from y to x decreases on average exponentially in the distance between x and y, independently of how many steps the QW may take

Properties

Proposition

Dynamical Localization $\Longrightarrow$ Spectral Localization, i.e. U_ω has pure point spectrum a.s.

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Defining the position operator $X(|x\rangle \otimes \tau) = |x|\ |x\rangle \otimes \tau$, we compute for an initial state

$$\mathbb{E}\left(\|X^{\frac{p}{2}} U_\omega^n \left(|0\rangle \otimes \tau\right)\|^2\right) = \sum_{|x\rangle \otimes e_j} |x|^p\ \mathbb{E}\Big(|\langle x \otimes e_j\,|\,U_\omega^n\,(0 \otimes \tau)\rangle|^2\Big) \leqslant \sum_n n^p\ C\,|S_n|\ e^{-gn}$$

The expectation is finite iff $|S_n| = o(e^{\gamma n})$ for some $\gamma < g$

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Fractional Moment Criterion

For all $s \in (0, \frac{1}{3})$ and $g > 0$ there exist $\varphi, c > 0$ such that for all $\sup_{x \in \Gamma} \|C_x - C_0\| \leqslant \varphi$ and $z \in \mathbb{C}$, $|z| \in (\frac{1}{2}, \frac{3}{2})$ we have

$$\mathbb{E}\Big(|\langle x \otimes \tau\,|\,(U_\omega - z)^{-1}\, y \otimes \tau'\rangle|^s\Big) \leqslant c\, e^{-g \mathrm{dist}(x,y)}$$

Correlated phases

Recall $D_\omega(|x\rangle \otimes e_j) = e^{i\omega_x}\ |x\rangle \otimes e_j$, we write D_ω in the ONB $\big(|x\rangle \otimes e_j\big)_{x\in\Gamma, j=1,2,3}$:

$$D_\omega = \begin{pmatrix} \ddots & & & & & \\ & e^{i\omega_x} & & & & \\ & & e^{i\omega_x} & & & \\ & & & e^{i\omega_x} & & \\ & & & & e^{i\omega_y} & \\ & & & & & e^{i\omega_y} \\ & & & & & & e^{i\omega_y} \\ & & & & & & & \ddots \end{pmatrix}$$

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Before: Fractional moment criterion required independent phases

$\longrightarrow$ generalization to correlated phases (ongoing work with A. Joye, S. Warzel)

Thank you for your attention!